Jinwon Sohn

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Education

- **Purdue University** — West Lafayette, USA
  Ph.D. - Statistics / Advisor: Prof. Qifan Song — *Jan 2021 - Aug 2025*
- **Yonsei University** — Seoul, Korea
  Master degree - Applied Statistics & Data Science / Advisor: Prof. Taeyoung Park — *Mar 2018 - Feb 2020*
- **Yonsei University** — Seoul, Korea
  Bachelor degree - Applied Statistics — *- Mar 2018*

Professional Career

- **University of Chicago Booth** — Chicago, USA
  Principal Researcher / Advisor: Prof. Veronika Ročková — *July 2025 - present*
  - **Roles**: Research in modern Bayesian statistics and machine learning, Mentoring Ph.D. students
- **Datarize** — Seoul, Korea
  Data Scientist — *2020*
  - **Roles**: Algorithm Evaluation, Developing Recommenders

Research Interest

- Synthetic Data Generation / Inference, Generative Modeling, Fairness-aware Machine Learning, Bayesian Statistics, Differential Privacy, Reasoning Large Language Models, Nonparametric Statistics, Causal Inference

Publication / In-press

***** : Co-first Author (or alphabetic order)

- **Sohn, J.** and Song, Q. (2026+). Parallelly Tempered Generative Adversarial Networks Toward Stabilized Gradients. *Journal of the American Statistical Association*, Special Issue on Statistical Science in Artificial Intelligence (in-press).
- **Sohn, J.**, Song, Q., Lin, G. (2026+). Task-tailored Pre-processing: Fair Downstream Supervised Learning. *Transactions on Pattern Analysis and Machine Intelligence* (in-press).
- **Sohn, J.**, Song, Q., Lin, G. (2024). Fair Supervised Learning with A Simple Random Sampler of Sensitive Attributes. *International Conference on Artificial Intelligence and Statistics* (pp. 1594-1602). PMLR.
- Kang, T., Kim, S., **Sohn, J.***, Awan, J. (2024). Differentially Private Topological Data Analysis. *Journal of Machine Learning Research.*
- **Sohn, J.***, Jeong, S., Cho, Y. M., Park, T. (2023). Functional Clustering Methods for Binary Longitudinal Data with Temporal Heterogeneity. In *Computational Statistics & Data Analysis*, 185, 107766.

Preprint

- Lim, T., Nam, K., **Sohn, J.*** Monotone curve estimation via convex duality. preprint arXiv:2501.06975. *Journal of the American Statistical Association.* Under Major Revision.
- Koo, E., **Sohn, J.***, Lim, T. Statistical Matching via Schrödinger Bridge beyond Conditional Independence. Submitted to *Neurips 2026.*

Working Papers

- **Sohn, J.**, Ročková, V. Private Generative Bayesian Bootstrap
  - Keywords: Differential Privacy, Nonparametric Bayesian inference, Generative Posterior Sampler
- **Sohn, J.**, Ročková, V. Is your LLM a Bayesian?
  - Keywords: Large Language Model, Subjective Bayesian
- Jeong, S., **Sohn, J.**, Ročková, V. Nonparametric Bayesian Synthetic Control
  - Keywords: Nonparametric Bayesian Inference, Causal Inference, AI-Powered Bayesian Inference

Talk

- 2026 Spring Data Science Finance, Statistics at Purdue University, USA
  - Task-tailored Pre-processing: Fair Downstream Supervised Learning
- 2025 Advances in Generative Modeling, the Joint Statistical Meetings, USA
  - Parallelly Tempered Generative Adversarial Networks
- 2025 Spring Quantitative Methods Seminar, School of Business at Purdue University, USA
  - Monotone Curve Estimation via Convex Duality
- 2024 Fall Graduate Student Workshop in Statistics, Purdue University, USA
  - Parallelly Tempered Generative Adversarial Networks
- 2024 Methods for Feature Selection, the Joint Statistical Meetings, USA
  - Fair Supervised Learning with A Simple Random Sampler of Sensitive Attributes
- 2024 Spring Purdue Graduate Student Organization Seminar, Purdue University, USA
  - Fair Supervised Learning with A Simple Random Sampler of Sensitive Attributes
- 2019 Fall Conference of the Korean Statistical Society, University of Seoul, Korea
  - Variational Inference on Functional Clustering of Varying Coefficients
- 2018 Fall Conference of the Korean Statistical Society, Ewha Woman University, Korea
  - Functional Clustering Methods for Binary Longitudinal Data with Temporal Heterogeneity

Award and Honor

- 2024 - 2025 Ross Lynn Research Scholar Grant for Statistics at Purdue University, USA
- College of Science Graduate Student Travel Award for Spring 2024 at Purdue University, USA
- 2024 High Profile Student Award for Research in Statistics at Purdue University, USA
- Third Place Award for Presentation, 2019 Fall Conference of the Korean Statistical Society, Korea
- Best Poster Award, 2018 Fall Conference of the Korean Statistical Society, Korea
- Grand Prize in 2018 Big Contest, National Information Society Agency, Korea
- Third Place Award in 2016&2017 Big Contest, National Information Society Agency, Korea
- High Honors - 2017 Spring & Honors - 2016 Fall

Software

- **fvcc**: Functional Clustering Methods for Varying Coefficients, written by R

Teaching

- **Purdue University** — West Lafayette, USA
  Teaching Assistant — *Spring 2021 - Fall 2022*
  - **STAT 303**: Probability and Statistics for Business
  - **STAT 512**: Applied Regression Analysis
  - **STAT 517**: Statistical Inference
- **Yonsei University** — Seoul, Korea
  Teaching Assistant — *Feb 2018 - Feb 2020*
  - **STA 3124**: Stochastic Processes
  - **STA 3126**: Mathematical Statistics

Skill

- **Languages**: Python(Adv), R(Adv), SQL

Reference

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